Design for Laser Powder Bed Fusion 304

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Class Outline

L-PBF and DFAM
L-PBF Design and Production Software
L-PBF Part Production
L-PBF Materials
L-PBF Applications
Review: Design for L-PBF
L-PBF Methods
Material Considerations for L-PBF Operating Conditions
L-PBF Support Methods
L-PBF Post-Processing
L-PBF Design Advantages and Limitations
Review: L-PBF Design Considerations

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L-PBF and DFAM

Parts produced by additive manufacturing (AM) go through the same five basic steps as traditionally manufactured parts: conceptualization, planning, designing, testing, and production. Design for manufacturing (DFM) is a method that can help manufacturers optimize each step in a variety of ways. Design for additive manufacturing (DFAM) is a specific type of DFM used for AM processes, such as powder bed fusion (PBF).

Laser powder bed fusion (L-PBF), the most common PBF method, uses a laser beam to heat layers of powdered material in order to build a part. L-PBF manufacturing requires knowledge of and ability to use software, as well as an understanding of machine settings, parameters, and operating conditions. Part designers use computer-aided design (CAD) and generative design software to conceptualize parts and define part dimensions, determine material composition, and establish part quality requirements. L-PBF machine operators use specialized software on L-PBF machine interfaces to set machine operating parameters and run the L-PBF production process. Design for L-PBF allows part designers to enhance design concepts, improve part functionality, increase cost savings, and reap other benefits when compared to traditional DFM processes.

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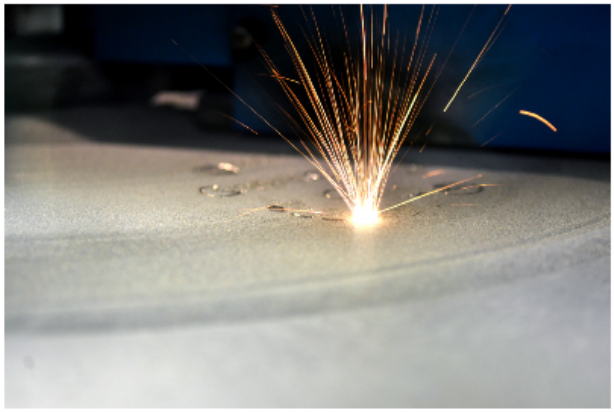

Figure 1:

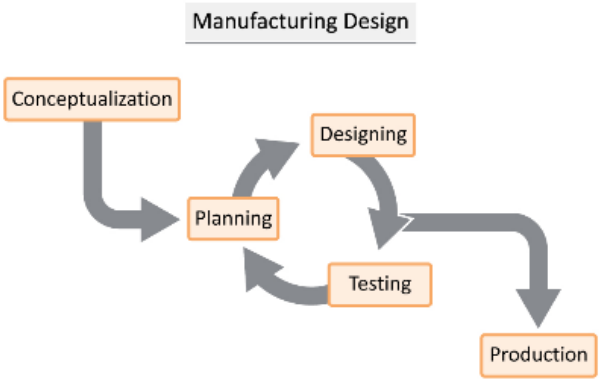

Figure 2:

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Lesson: 2/12

L-PBF Design and Production Software
Digital tools for L-PBF include CAD software for design and specialized software for running the production process.

In order to produce a part with L-PBF, a design engineer typically creates a 3D model of the part using CAD software. CAD software ranges from free software for basic design to more expensive software with more advanced design capabilities. Once the CAD model is complete, it is converted into stereolithography (STL) file format. While some AM methods translate the STL file into a traditional G code file, L-PBF uses specialized build processor software to produce a different file type. The build processor software, which is usually specific to the machine, converts the STL file into a job file that the L-PBF machine can read. The job file controls the laser temperature, the laser path, the powder application, and the movement of the machine based on the part geometry, material types, and machine settings.

Video content available online.

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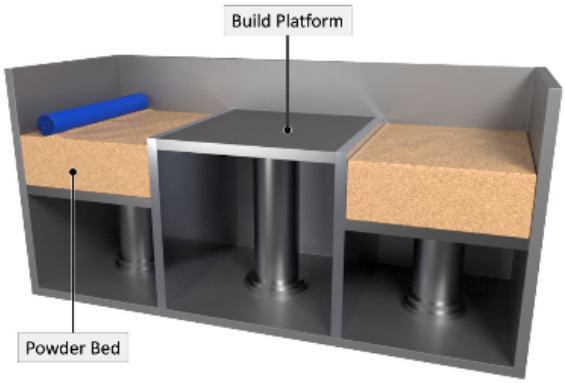
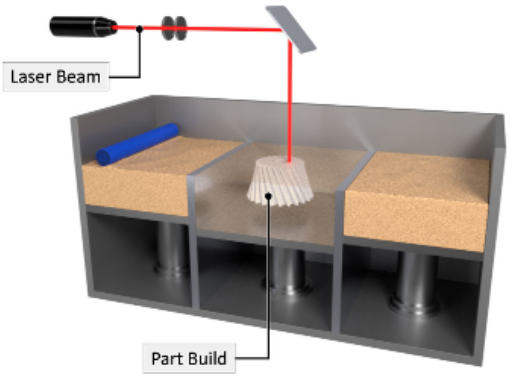
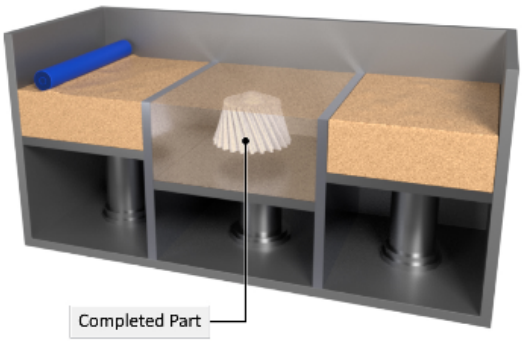
L-PBF Part Production



Before the L-PBF production process can begin, the powder bed must be filled with powder and loaded onto the L-PBF machine. Most machines then feed the powder upward from the bed into the build chamber. The powder is fed in specific quantities to form each layer, and the powder blade spreads each new powder layer out on the build platform.



The laser beam is emitted through a lens to fuse or melt precise areas of each layer of powder. The specific temperature of the laser depends on the type of material. The path of the laser is controlled by specialized software in the L-PBF machine to fuse or melt areas of each powder layer based on part dimensions specified in the original CAD model of the part.



As each layer of the part is completed, the build platform lowers the part deeper into the build chamber. The part grows until it is fully submerged in the unused powder in the build chamber. Once the entire part is complete, the part is typically raised to the top of the build chamber and removed by an operator. On some machines, the powder bed is removed first, and the powder is then emptied out before the completed part is removed.

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completed part is removed.

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L-PBF Materials

Laser powder bed fusion processes can use a wide range of materials, including polymers, ceramics, and metals. Powdered metal is the most common type of material used in L-PBF applications and is available in a much wider selection than polymers or ceramics. Metals are strong and have the highest electrical and thermal conductivity. L-PBF make parts from pure metals, metal alloys, and metal matrix composites (MMCs). Producing metal parts with L-PBF allows for greater design possibilities than machining since cutting tools provide less flexibility than building a part from scratch.

L-PBF also uses polymers that can be processed into powder form, including most thermoplastics. Most polymers are ductile and have low electrical conductivity and high chemical resistance. Low conductivity makes polymers useful insulators, while chemical resistance improves their ability to withstand corrosion. Polymer is often used for rapid prototyping but also can be used for end-use parts.

Powdered ceramics can also be used in L-PBF applications. Ceramics are hard and brittle. They provide parts with very high heat resistance and low electrical conductivity. Glass is the most common ceramic material used in L-PBF, but other materials, including carbon fiber, can also be used. Some ceramic parts processed with L-PBF include a binder material, such as nylon, to help fuse the ceramic particles together.

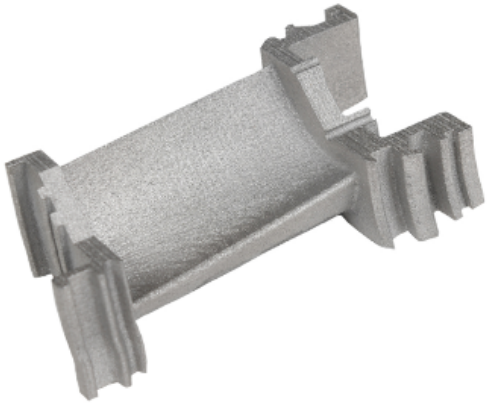

Figure 1: Metal Turbine Blade, courtesy of Xact Metal

Figure 2: A range of materials can be processed into powder form.

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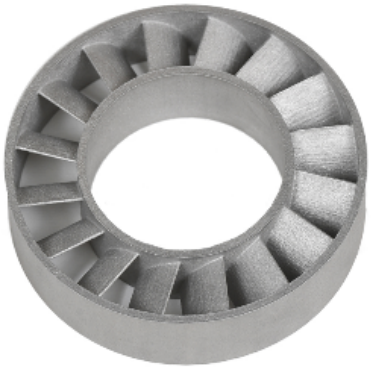
L-PBF Applications



Stator for Aerospace, Courtesy of Xact Metal

Like other metal AM methods, L-PBF is most often used for aerospace components. L-PBF can create parts with highly complex geometry and parts made of various metals and alloys that combine desired properties, such as ductility and strength, in a single assembly.

Car Bumper

Many automotive part manufacturers have adopted powder bed fusion to produce unique parts such as car bumpers, exhaust manifolds, and other components. L-PBF enables more design options to improve functionality, faster design-to-production speeds, and greater flexibility for mass customization than traditional manufacturing methods.

Knee Replacement Model

The medical equipment industry uses L-PBF for a range of production purposes. L-PBF can use ceramics to develop dental implants, knee and hip replacements, and other bone structures. Some specialized L-PBF applications also use biomaterials for tissue engineering.

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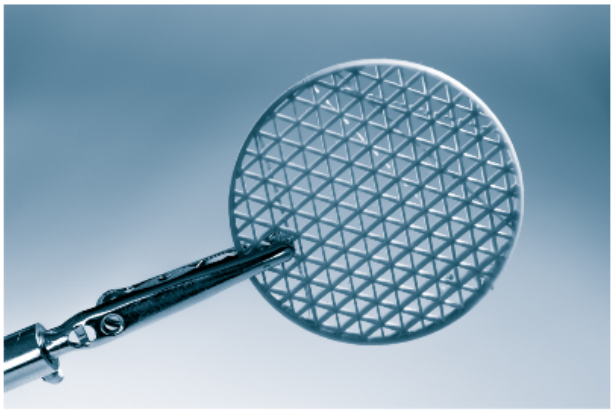

Lattice Structure

Manufacturers can use L-PBF to improve the geometrical design of components to add functionality. For example, adding conformal cooling channels improves heat management. L-PBF can also be used for lightweighting by creating honeycomb structures or lattice structures in the part's interior. This reduces the amount of material. When produced by traditional manufacturing processes, the same type of part would typically be completely solid.

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Lesson: 6/12

Review: Design for L-PBF

1. Design for additive manufacturing follows a different set of process steps than traditional design for manufacturing. **FALSE**
2. In a laser powder bed fusion process, the part is lowered on the build platform as each layer of the part is built. **TRUE**
3. Laser powder bed fusion is sometimes used to build medical devices from ceramic material. **TRUE**
4. Most powder bed fusion production software uses a basic G code file, similar to traditional CNC production software. **FALSE**
5. Laser powder bed fusion systems can use multiple material types. **TRUE**

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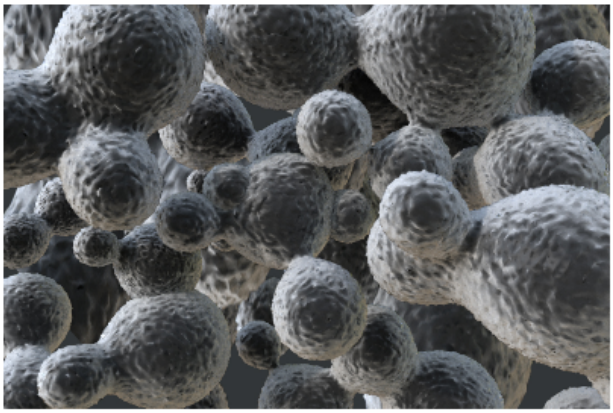
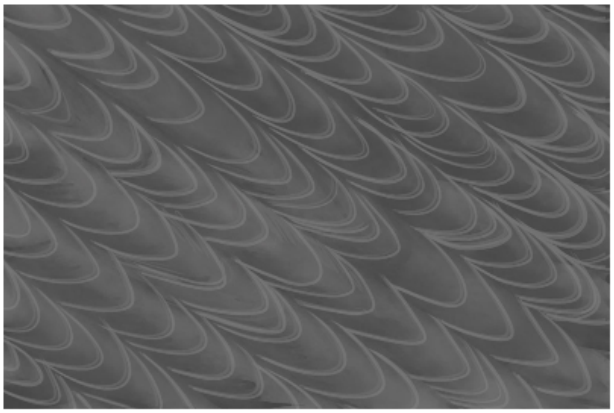
L-PBF Methods



SLS fuses layers, leaving some porosity.

In selective laser sintering (SLS), the powdered material is heated to a temperature below its melting point that allows the powder particles to fuse together. The SLS process can sinter polymers, ceramics, or different types of metal alloys together by adjusting laser temperatures. When used with powdered metals, SLS is often referred to as direct metal laser sintering (DMLS). Because it uses lower temperatures, SLS produces little or no heat distortion and can create strong, highly accurate builds. However, SLS part builds typically have some porosity between fused layers.



SLM melts layers, eliminating porosity.

Selective laser melting (SLM) heats the powdered material above its melting point. Unlike SLS, SLM completely melts the material to create a single, solid-state part with no porosity between powder particles. SLM is most practical when using a single type of material since using alloys or multiple materials involves more than one melting point. Since it fully melts the material, it can produce high-performance parts that are stronger and denser than parts produced with SLS. However, fewer types of material can be used with SLM than SLS. It also uses more laser energy than SLS, which can cause more heat distortion.

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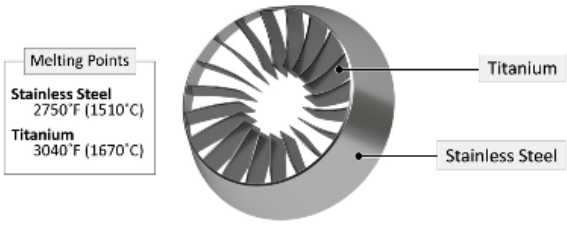
Material Considerations for L-PBF Operating Conditions



In both SLS and SLM, each material requires different temperature settings and other parameters that design engineers and operators must consider. Thermal stress from the laser heat source is a primary consideration for L-PBF design and production processes. Designers and manufacturers must use appropriate heat settings for different materials to prevent defective parts. SLS processes' lower temperatures create less thermal stress than SLM. However, many of the alloys and MMCs used with SLS contain materials with different melting points. The laser must provide enough heat to fuse materials with higher melting points, which may cause some thermal stress for materials with lower melting points. When using SLM, designers and operators must account for more thermal stress and more potential inaccuracy than with SLS due to the higher operating temperatures. Since the SLM process completely melts the material, expansion and shrinkage typically occur as the metal melts, then cools back into solid form. While there is little or no distortion with SLS, some powder may not fuse properly during the SLS process, and parts may also have porosity, which is not a concern with SLM.

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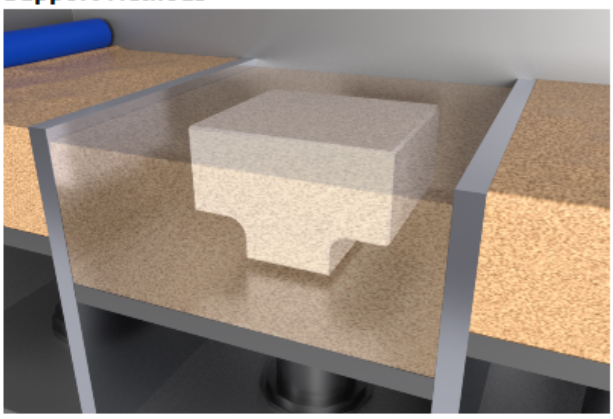
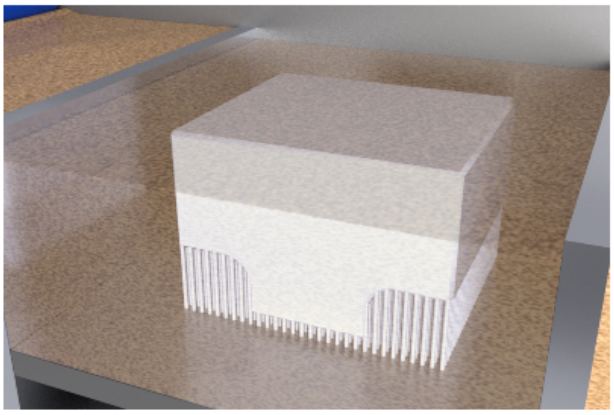
L-PBF Support Methods



Passive Support

L-PBF processes, especially SLS, often use the loose powder in the bed itself to support free-standing parts and overhangs instead of using support structures. This is sometimes called the passive support method. Passive support provides advantages such as enabling complex build shapes, reducing material costs, and allowing multiple parts to be produced in one build cycle. However, using passive support instead of a build platform may result in more powder being heated but not completely fused. The material properties of this unfused powder may be altered, which can create problems with recycling the powder.



Active Support

Some L-PBF applications, particularly high-temperature SLM, use support structures in order to prevent part distortion that can occur during the build process. Building support structures onto the build platform to support overhangs and other features is often called the active support method. Support structures provide a base to allow better heat flow during the build process, which can reduce the amount of thermal stress on the material used in the build. Manufacturers must remove support structures using cutting tools and often use computer numerical control (CNC) machines for this. Part designers must consider how the build will be oriented on the CNC machine when designing parts that use support structures.

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L-PBF Post-Processing

Like other AM processes, L-PBF usually requires post-processing operations once the part is complete. L-PBF processes, particularly SLS, produce parts with greater accuracy than extrusion-based systems. L-PBF can build parts to net shape. However, SLS parts require finishing processes to remove loose, unfused powder and correct poor surface finish. These finishing processes may require grinders, end mills, and other finishing tools. Since SLS produces parts with some porosity between particles of the build material, other processes, like infiltration sintering, may also be necessary. The infiltration process melts a filler metal into the finished build in order to fill in pores and increase the part's density.

SLM methods may require cutting tools to remove support structures once the build is completed. Additionally, the higher temperatures used in SLM create wider temperature gradients. Temperature gradient measures the rate of temperature change with distance in a given direction. Wider gradients on a part build mean more extreme temperature changes, which can result in greater distortion and shrinkage of the material particles. Distortion and shrinkage can cause permanent changes that make part dimensions less accurate. For this reason, parts produced by SLM may require heat treatment after or even partway through the build process to reduce thermal damage, and finishing tools to machine the completed part down to its original CAD dimensions in order to meet high tolerances.

Figure 1: Abrasive Finishing

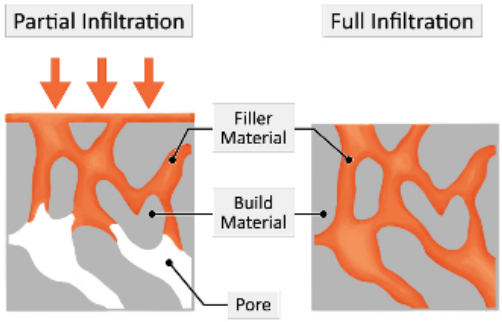

Figure 2: Infiltration sintering increases part density.

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L-PBF Design Advantages and Limitations

Compared to most other AM methods, L-PBF excels at creating highly accurate, complex builds. SLS in particular creates highly accurate dimensions due to little or no heat distortion . SLS's accuracy is similar to the accuracy of binder jetting, an AM method that fuses parts at room temperature, but SLS can also build stronger parts with greater durability than binder jetting. L-PBF is also one of few AM processes that can print metal parts for end use. DMLS specifically allows manufacturers to build parts made of various metals and alloys.

While L-PBF enhances design possibilities, it also has limitations that are important for designers to consider. Most L-PBF machines have a limited build size and cannot apply different materials to one build as easily as extrusion methods. This can result in slower build times for parts designed with multiple materials. With L-PBF, one powdered material must be removed from the powder bed before another material can be used in the process. L-PBF also requires a sealed, low-oxygen chamber in order to prevent oxidation and contamination, while metal extrusion may only require a shielding gas.

In addition, L-PBF has greater up-front costs than other AM methods like directed energy deposition (DED), which melts and deposits powdered metal through a small opening in a nozzle, because L-PBF requires more powder. However, since unused powder can be recycled, effective project planning can achieve savings over the long term.

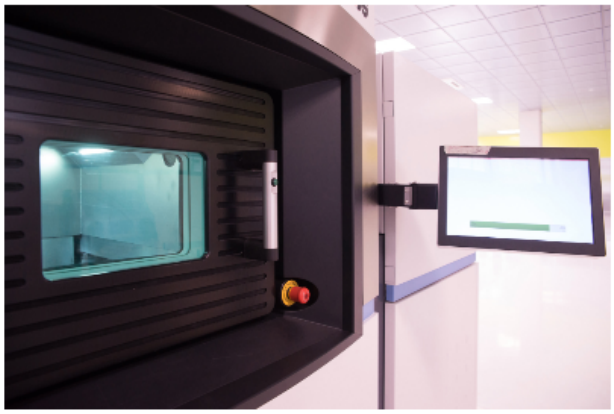

Figure 1: Build Chamber

Figure 2: Complex Build

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Lesson: 12/12

Review: L-PBF Design Considerations
Which of the following is a strength of laser powder bed fusion (L-PBF)?

- **High dimensional accuracy**
- Faster build rates
- Temperature gradients
- Support structures

What is the most common type of material used in laser powder bed fusion (L-PBF)?

- **Metal**
- Ceramic
- Polymer
- Composite

Which of the following can be improved by using active support instead of passive support?

- **Heat flow**
- Geometrical complexity
- Dimensional accuracy
- Build rate

Which of the following is more likely to be a concern in selective laser melting processes than in selective laser sintering processes?

- **Thermal stress**
- High porosity
- Build rate
- Geometry limitations

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Class Vocabulary

Active Support	A method that uses support structures to support certain areas of a part during the powder bed fusion process. Active support can help control the flow of heat through certain areas of a part to prevent heat damage.
Additive Manufacturing	AM. The process of joining or solidifying materials to make an object from a three-dimensional computer model. Additive manufacturing methods typically build up layers of material to create an object.
Alloys	A material created by intentionally mixing two or more other materials, one of which must be metal. Alloys often have the desirable mechanical and physical properties of both materials.
AM	Additive manufacturing. The process of joining or solidifying materials to make an object from a three-dimensional computer model. AM methods typically build up layers of material to create an object.
Binder	A substance that holds together two or more other materials. A binder is used to hold together powdered metal to form a solid part in binder jetting additive manufacturing (AM) processes.
Binder Jetting	An additive manufacturing (AM) method in which liquid binder is used to join powdered materials to create parts. Binder jetting can create parts out of metal, polymer, or ceramic.
Biomaterials	A natural material that composes a part or all of a living structure. Biomaterials, like living cells and tissues, can be used in some additive manufacturing methods.
Build Chamber	The area contained within an additive manufacturing (AM) machine where the part is actually built. Many build chambers are enclosed to improve the safety of the manufacturing operation.
Build Platform	The flat surface on which a part is additively manufactured. The build platform can either be a permanent machine surface from which parts are removed or a surface that can be removed from a machine once the build is complete.
Build Processor	A type of software used in powder bed fusion (PBF) or other additive manufacturing (AM) machines that converts STL files into job files. Build processor software is usually specific to each machine.
CAD	Computer-aided design. A method of designing two- and three-dimensional objects using computers and software. CAD is most often used to create part models for production.
Ceramics	A hard, brittle material that can withstand high temperatures and resist corrosion. Ceramic parts can be created through additive manufacturing (AM) processes such as binder jetting and selective laser sintering (SLS).
Chemical Resistance	The ability of a material to resist chemical changes. Chemical resistance prevents gradual degradation caused by the atmosphere, moisture, or other substances.
Computer Numerical Control	CNC. A combination of software and hardware that directs the operation of a machine. CNC machines are used for subtractive manufacturing.
Computer-Aided Design	CAD. A method of designing two- and three-dimensional objects using computers and software. Computer-aided design is most often used to create part models for production.
Conceptualization	The process of forming an idea about a possible part or manufactured good. Conceptualization is one of the initial stages of design for manufacturing (DFM).
Conformal Cooling	Cooling channel configurations that curve and closely correspond to a part's shape. Conformal cooling provides better heat management and helps reduce tool wear and cycle time.
Design For Additive Manufacturing	DFAM. The methodology of planning, testing, and creating an additively manufactured part that functions optimally. Design for additive manufacturing allows engineers to mostly focus on part functionality.
Design For Manufacturing	DFM. The methodology of planning, testing, and creating a part that functions correctly and is easy to manufacture. Design for manufacturing, also known as design for manufacturability, involves considering part functionality and the limits of the manufacturing process.

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considering part functionality, and the limits of the manufacturing process.

Designing The process of creating the actual specifications for a potential part. Designing involves creating the blueprints and prototypes for a part.

Direct Metal Laser Sintering DMLS. A powder bed fusion (PBF) process that builds a metal part by heating the powdered metal to just below its melting point using a laser. Direct metal laser sintering can produce complex metal parts, but they often require extensive post-processing.

Directed Energy Deposition DED. An additive manufacturing (AM) process in which focused thermal energy is used to melt materials as they are fed or blown through a nozzle. Directed energy deposition is often used with powdered or wire metal.

DMLS Direct metal laser sintering. A powder bed fusion (PBF) process that builds a part by using a laser to heat and fuse together successive layers of powdered metal. DMLS can produce complex metal parts, though they often require extensive post-processing.

Ductile Able to be drawn, stretched, or formed without breaking. Ductile materials generally have low strength.

Ductility The ability of a material to be drawn, stretched, or formed without breaking. Ductility is more common in softer materials.

Electrical Conductivity The ability of a material to convey electricity. Metals such as aluminum and copper have high levels of electrical conductivity.

End Mills A milling cutter used to machine grooves, slots, pockets, and contours into workpieces. End mills have one or more flutes.

End-Use Designed to be used directly by a consumer or directly in another manufactured product. End-use products created by AM include medical implants, custom dental devices, and camera equipment.

Extrusion A type of additive manufacturing process where raw material is deposited from a nozzle or other application tool. Extrusion methods include fused deposition modeling and material extrusion.

Finishing A final machining process that achieves the desired surface finish and tolerance for a part or component. Finishing is often performed on grinders or with cutting tools.

G Code A method of programming that pairs address letters with numerical values to form words. G code programs are used in additive manufacturing, CNC machining, and hybrid machining.

Generative Design An advanced type of computer-aided design (CAD) that uses cloud computing and artificial intelligence to design products. Generative design software can allow engineers to innovate designs and produce alternative designs much faster than traditional CAD software.

Geometry The measurement, properties, and relationships of the lines and points of an object that make up its shape. Geometry is used when measuring and designing parts.

Heat Resistance The ability to resist degradation when under extreme heat. Heat resistance is important for materials that will be exposed to heat on a regular basis.

Heat Treatment A controlled heating and cooling process used to change a material's structure and alter its physical and mechanical properties. Heat treatment often alters a material's hardness.

Honeycomb Structures A type structure that consists of empty cells separated from each other by a solid wall. Honeycomb structures can have square or hexagonal cells.

Infiltration Sintering The process of melting a filler material into part with a higher melting temperature in order to fill any empty spaces or voids in the part. Infiltration sintering is a specialized post-processing step that is sometimes used to harden and strengthen parts produced by selective laser sintering.

Insulators A material or element that has little electrical conductivity and high resistance to electrical charges. Most insulators are plastics and ceramics.

Interfaces A control panel and display that an operator uses to interact with and control a machine. Interfaces are unique to each additive manufacturing (AM) machine and require operator training to use.

Job File A program created using build processor software that controls the functions of the laser and other parts of the powder bed fusion machine. Job file formats may be specific to each machine.

Laser Beam An intense beam of coherent, collimated, monochromatic light that can be precisely aimed and controlled. Laser beams are used in laser powder bed fusion (L-PBF) and other additive manufacturing (AM) processes.

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manufacturing (AM) processes.

Laser Powder Bed Fusion L-PBF. An additive manufacturing (AM) method that uses a laser to fuse or melt together layers of powdered polymer, metal, ceramic, or other material. Laser powder bed fusion processes include selective laser sintering (SLS) and selective laser melting (SLM).

Lattice Structures A repeating, symmetrical pattern of crossing strips of material that leaves diamond- or square-shaped gaps between them. Lattice structures provide excellent strength to a part.

L-PBF Laser powder bed fusion. An additive manufacturing (AM) method that uses a laser to fuse or melt together layers of powdered polymer, metal, ceramic, or other material. L-PBF processes include selective laser sintering (SLS) and selective laser melting (SLM).

Manifolds A fluid conductor that consists of a metal block with multiple integrated channels. Manifolds are designed for use with multiple control valves or to conduct fluid in confined spaces.

Mass Customization The ability to quickly create a large number of uniquely designed variations on a part. Mass customization is one of the key advantages of additive manufacturing (AM) technology.

Metal Alloys A material created by intentionally mixing two or more other materials, one of which must be metal. Metal alloys often have the desirable mechanical and physical properties of both materials.

Metal Matrix Composites MMC. A composite made from a metallic matrix and high-performance reinforcement materials. Metal matrix composites have high levels of strength and stiffness, and relatively high resistance to heat.

MMCs Metal matrix composite. A composite made from a metallic matrix and high-performance reinforcement materials. MMCs have high levels of strength and stiffness, and relatively high resistance to heat.

Net Shape A part that meets consumer or manufacturer specifications, including tolerance and surface finish, directly after it is manufactured. Net shape parts do not need any further post-processing after they are created.

Nylon A semicrystalline thermoplastic that gains strength when the fibers are stretched. Nylon is a commonly used term for polyamides (PA).

Oxidation A chemical reaction involving the addition of oxygen, the removal of hydrogen, or the removal of electrons from a material. Oxidation weakens and degrades a material.

Passive Support A method that uses the powder in the powder bed as support rather than using support structures. Passive support allows technicians to build multiple parts in a single process.

Planning Assessing the ways to best create a conceptualized part. Planning is a phase in the design process that involves initial considerations of the part design and practical concerns related to the manufacturing process.

Polymers A lightweight material that generally has high corrosion resistance, a high strength-to-weight ratio, and a low melting point. Polymers, or plastics, are some of the most common manufacturing materials.

Porosity The presence of small spaces or voids within a solid material. Porosity in a material can potentially affect the durability of the part.

Post-Processing A finishing process used to clean, grind, or otherwise prepare an additive manufacturing part for shipping to a manufacturer or other consumer. Common additive manufacturing post-processing steps include removing support structures, heat treating, improving surface finish, and bringing the part into tolerance.

Powder Bed The area of a powder-based additive manufacturing (AM) machine that holds granulated build material and provides or holds the build platform. Powder beds are used in powder bed fusion (PBF) and binder jetting.

Powder Bed Fusion PBF. An additive manufacturing (AM) process that uses adhesives, heat, or light to bond areas in a container of powdered polymer, metal, ceramic, or other material. Powder bed fusion processes include binder jetting, selective laser sintering (SLS), and direct metal laser sintering (DMLS).

Powder Blade A device that slides forwards and backwards over the surface of a powdered material in a powder bed. Powder blades are used in laser powder bed fusion (L-PBF) systems to smooth powdered material layers over top a part or build platform.

Production The manufacturing of a finished part for delivery to a customer. Production is the final stage of design for manufacturing (DFM).

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Rapid Prototyping A product development technique that uses additive manufacturing methods to create a part model. Rapid prototyping allows manufacturers to quickly create functional part models in a short time period, reducing lead time.

Selective Laser Melting SLM. A powder bed fusion (PBF) process that builds a part by using a laser to melt together successive layers of powdered materials. Selective laser melting is similar to selective laser sintering (SLS) but produces stronger and denser parts by fully melting build materials.

Selective Laser Sintering SLS. A powder bed fusion (PBF) process that builds a part by using a laser to heat and solidify layers of powdered material. Selective laser sintering produces parts that are strong and durable but usually have poor surface finish.

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Stereolithography STL. An additive-manufacturing (AM) compatible file format that represents 3D models as a series of interconnected triangles. STL files are sometimes referred to as stereolithography files, standard tessellation language files, or standard transform language files.

Strength The ability of a material to resist forces that attempt to break or deform it. Increased strength can make materials more difficult to machine.

Support Structures A reinforcing component used to hold the weight of an additively manufactured part as it is being constructed. Support structures are removed from the part once the build is complete.

Temperature Gradients The rate of temperature change measured over a distance. Temperature gradients across the surface of a part during production can affect its material quality.

Testing The examination of a part to ensure that it performs its intended function and that it can be satisfactorily manufactured. Testing is a design step that indicates whether the part needs additional planning or if it is ready for production.

Thermal Conductivity The ability of a material to transfer heat energy through it. Materials with low thermal conductivity make good heat insulators.

Thermal Stress Damage to a material due to excessive changes in temperature. Thermal stress can warp or distort a part.

Thermoplastics A grouping of plastics that can be repeatedly reheated and re-shaped. Thermoplastics are the most commonly used plastics.

Tissue Engineering The use of biomaterials and other materials to rebuild biological tissues. Tissue engineering processes, which can restore tissues in the human body, can be done using some additive manufacturing (AM) methods.

Tolerances An unwanted but acceptable deviation between an actual part dimension and its intended design. Tolerances are assessed during the inspection stage of manufacturing.

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